```
%simpson 1/3 rule of integration

function[I]=simpson13(f,a,b,n)
if mod(n,2)==1
    disp('please enter an even number')
    return
end
h=(b-a)/n
if numel(f)>1
    I=(h/3)*((f(1)+f(n+1))+4*sum(f(2:2:(n)))+2*sum(f(3:2:n-1)))
else
    x=a:h:b
     I=(h/3)*((f(a)+f(b))+4*sum(f(x(2:2:n)))+2*sum(f(x(3:2:(n-1)))))
end
end
```

```
%to find the integral of the function using simpson1/3 rule
simpson13(@(x) exp(-x.^2),0,0.6,6)
```

```
h =
0.1000
x = 1×7
         0    0.1000    0.2000    0.3000    0.4000    0.5000    0.6000
I =
0.5352
ans =
0.5352
```

```
simpson13(@(x) 1\(1+x.^2),0,1,4)
```

```
h =
0.2500
x = 1×5
         0    0.2500    0.5000    0.7500    1.0000
I =
1.3333
ans =
1.3333
```

```
simpson13(@(x) 1\log10(x),2,8,6)
```

```
h =
1
x = 1×7
     2     3     4     5     6     7     8
I =
4.0164
ans =
4.0164
```

```
simpson13(@(x) (1-8*x.^3).^(3/2),0,0.3,6)
```

```
h =
0.0500
```

```
x = 1×7
        0    0.0500    0.1000    0.1500    0.2000    0.2500    0.3000
I =
0.2765
ans =
0.2765
```